The Definition of the Derivative

ANSWER KEY

Using the limit definition

1 Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.

$\frac{3(x+h)^2 - (x+h) - (3x^2 - x)}{h} = \frac{6xh + 3h^2 - h}{h} = 6x + 3h - 1 \to 6x - 1.$

2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.

$\frac{\frac{1}{x+h+2} - \frac{1}{x+2}}{h} = \frac{(x+2) - (x+h+2)}{h(x+h+2)(x+2)} = \frac{-h}{h(x+h+2)(x+2)} = \frac{-1}{(x+h+2)(x+2)} \to \frac{-1}{(x+2)^2}.$

The derivative as a limit at a point

3 Use $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.

$f(3+h) = (3+h)^2 - 2(3+h) = 9 + 6h + h^2 - 6 - 2h = 3 + 4h + h^2$. $f(3) = 3$. $\frac{4h + h^2}{h} = 4 + h \to 4$: $f'(3) = 4$.

4 Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

$\frac{x^3 - 8}{x - 2} = \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = x^2 + 2x + 4 \to 4 + 4 + 4 = 12$: $f'(2) = 12$.

Differentiability and graphical interpretation

5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.

Left-hand: $\lim_{h \to 0^-} \frac{|h|}{h} = -1$. Right-hand: $\lim_{h \to 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(2)$ does not exist (the graph has a corner).

6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).

(1) A corner — one-sided derivatives disagree, as with $|x|$. (2) A vertical tangent line — the slope becomes infinite, as with $\sqrt[3]{x}$ at $x = 0$. (3) A discontinuity — if f isn't continuous at a point, it can't be differentiable there, since differentiability implies continuity.

Power, sum, and constant rules

7 Differentiate each function using the power rule:

a $f(x) = 7x^5 \quad = 35x^4$ **b** $f(x) = \frac{4}{x^2} \quad = -8x^{-3}$ **c** $f(x) = 6\sqrt[3]{x} \quad = 2x^{-2/3}$ **d** $f(x) = 9 \quad = 0$

8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.

$f'(x) = 20x^3 - 9x^2 + 2.$

Tangent lines from the derivative

9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.

$f(1) = 3 - 1 = 2$; $f'(1) = 6 - 1 = 5$. Tangent line: $y - 2 = 5(x - 1)$, i.e. $y = 5x - 3$.

10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).

$f(2) = 8$. Tangent line: $y - 8 = 12(x - 2)$, i.e. $y = 12x - 16$.

Estimating derivatives numerically

11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.

$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{3.98 - 3.42}{0.2} = \frac{0.56}{0.2} = 2.8$.